

Ontology Population Using LLMs: Which Factors Matter? (Appendix)

Appendix 1: Formalization of Evaluation Metrics

For each individual, the ground truth is defined as the sequence of concepts, starting at the individual's directly asserted concept, along the path to the top concept of that ontology. We provide a mathematical formulation of mAP@K with $K = 1, D$ being applied to obtain mAP@1 and mAP@D respectively.

Let the ground truth hierarchy for an individual t be given by:

$$\{c_i \mid c_i \in C ; c_i \subseteq c_{i+1} ; t \in \Sigma_{c_1} ; 1 \leq i \leq D - d + 1\} \quad (1)$$

where d is the depth of the directly-asserted concept c_1 and $d \leq D$

Let $\{y_i\}_{i=1}^{D-d+1}$ be the predicted hierarchy for t

The mAP at a sequence length of $K(\leq D - d + 1)$ is given by:

$$\text{mAP@K} = \frac{1}{|T|} \sum_{t \in T} \text{AP@K}_t \quad (2)$$

where,

T is the set of individuals;

$|\cdot|$ is set cardinality and;

AP@K _{t} is the average precision for t at a sequence length K , given by:

$$\text{AP@K}_t = \frac{1}{K} \sum_{x=1}^K \frac{|\{y_i\}_{i=1}^x \cap \{c_j\}_{j=1}^x|}{|\{y_k\}_{k=1}^x|} \quad (3)$$

The numerator in Equation 3 is the cardinality of the set of *true positives* and the denominator is the cardinality of the set of *predictions* for a single individual t .

Appendix 2: String Similarity Prediction

Figure 1 highlights the hierarchy predicted by GPT-4o for an individual ‘AccountFacet’ in the CASE ontology on GPT-4o under different zero and few-shot conditions. In the zero-shot scenario, GPT-4o predicts primarily siblings rather than ancestors based on strong string similarity. As the number of examples provided increases, GPT-4o is able to predict the correct hierarchy.

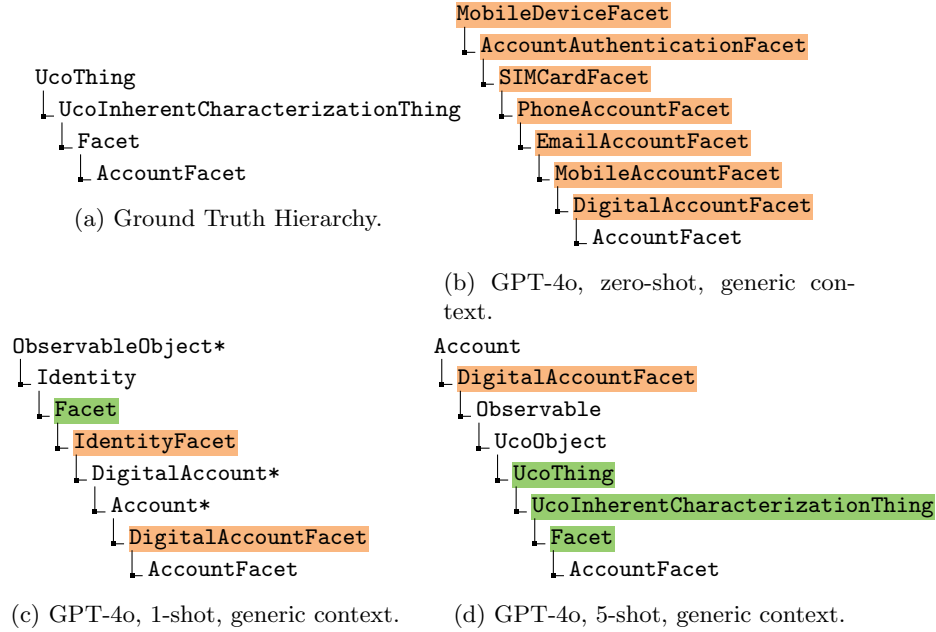


Figure 1: Ground Truth and predicted hierarchies of ‘AccountFacet’ taken from individual correctly identified as instance of ‘AccountFacet’. Concepts in orange are siblings of ‘AccountFacet’ (incorrect hierarchy). Concepts in green are correctly identified ancestors. * are incorrect ancestors but belong to the correct hierarchy.